Lab 4 DMM Voltage and Current Measurements and Voltage-Divider Circuits

1.0 Introduction

In this lab, students will apply their knowledge of the DC power supply, DMM, and breadboard to build and test a two-resistor voltage-divider circuit.

Students will measure DC voltage and current, connect the DMM correctly as a voltmeter and ammeter, and compare measured results with theoretical values. The effects of resistor tolerance and measured resistance on circuit voltage and current will also be examined.

1.1 Schematics and Connection Diagrams

A voltage divider consists of two resistors connected in series across a voltage source, as shown in Figure 1.1. The same current flows through both resistors, and the input voltage V_{IN} is divided into the voltage drops V_1 and V_2.

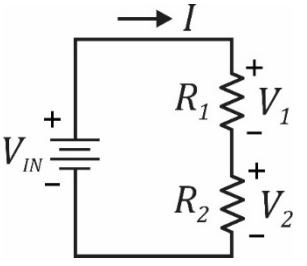


Figure 1.1 Simple two-resistor voltage-divider circuit

1.2 Equations

1.2.1 Ohm's Law

$$V = IR$$

1.2.2 Kirchhoff's Voltage Law (KVL)

$$V_{IN} = V_1 + V_2$$

1.2.3 Voltage Divider

$$V_1 = V_{IN}\frac{R_1}{(R_1 + R_2)}$$

$$V_2 = V_{IN}\frac{R_2}{(R_1 + R_2)}$$

1.2.4 Resistor Tolerance

A resistor's actual resistance may differ from its nominal value, R_N. For tolerance T, expressed as a decimal, the allowable range is

$$R_{min} = R_N(1 - T)$$
$$R_{max} = R_N(1 + T)$$

For example, a $100\ \Omega$, ±5% resistor may range from $95\ \Omega$to $105\ \Omega$. Therefore, measured voltages and currents may differ slightly from values calculated using nominal resistances.

2.0 Required Equipment, Parts and Tools

- Digital Multimeter, Keysight 34450A
- DC Power Supply, Keysight EDU36311
- 1.0 kΩ Resistor (1/2 W, 5%), quantity 1
- 1.5 kΩ Resistor (1/2 W, 5%), quantity 1
- Breadboard
- Test Leads; banana to alligator, red and black, quantity 2 each

3.0 The Voltage Divider Circuit

The circuit shown in Figure 3.0 consists of a 1.0 kΩ resistor and a 1.5 kΩ resistor connected in series across a 10 V source. The same current flows through both resistors, and the larger resistor has the larger voltage drop.

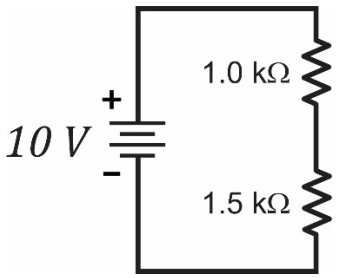


Figure 3.0 Voltage-divider circuit used in this experiment

3.1 Identify the Voltage-Divider Resistor

ACTION: Collect one 1.0 kΩ resistor and one 1.5 kΩ resistor, each with a 5% tolerance. Use the resistor color bands shown in Figure 3.1 to identify the resistance value and tolerance.

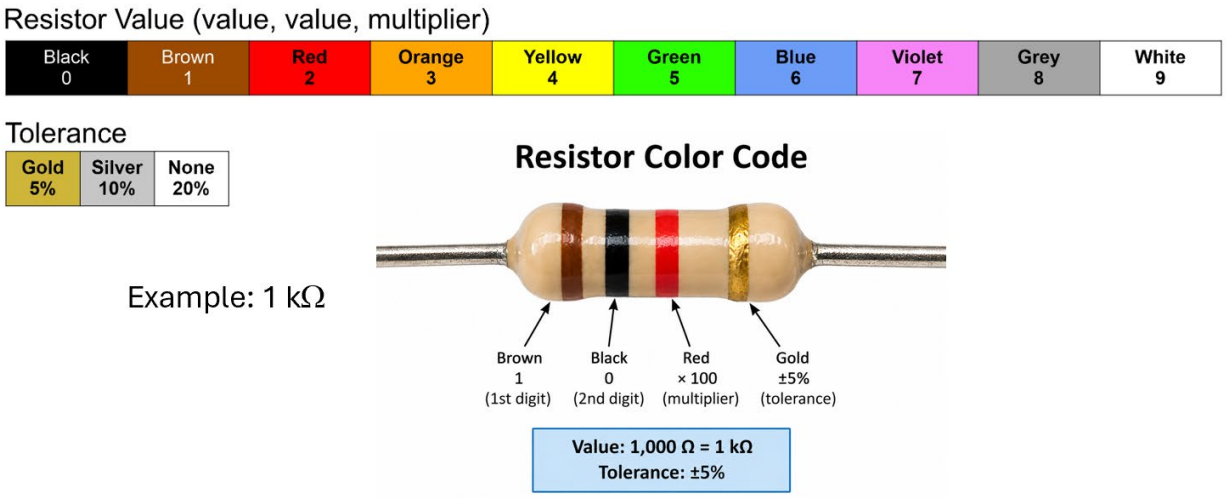


Figure 3.1 Resistor color bands used to identify resistance value and tolerance

QUESTIONS: On the handout, write your answers to the questions listed under Section 3.1

3.2 Measuring R1 and R2

ACTION: Configure the DMM as an ohmmeter. Measure and record the resistance of each the 1.0 kΩ and 1.5 kΩ resistors.

QUESTIONS: On the handout, write your answers to the questions listed under Section 3.2

3.3 Configure the 10 V Source

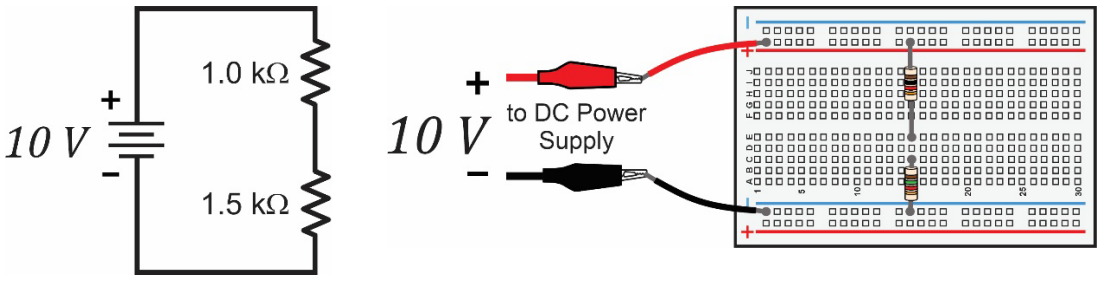
ACTION: Configure one power-supply channel for $+10$ Vin constant-voltage (**CV**) mode. Set the **current limit at $10\ mA$.**

SAFETY: For CV mode, the current limit should be set higher than the expected current, in this case, above 4 mA.

SAFETY: After setting the DC power supply voltage, turn off the power supply channel until you are ready to make a measurement. This reduces the chance of accidentally shorting the power supply while the circuit is being assembled or modified.

3.4 Assemble the Voltage-Divider Circuit

ACTION: Assemble the voltage-divider circuit on the breadboard as shown in Figure 3.4. Do not turn on the power-supply channel output.

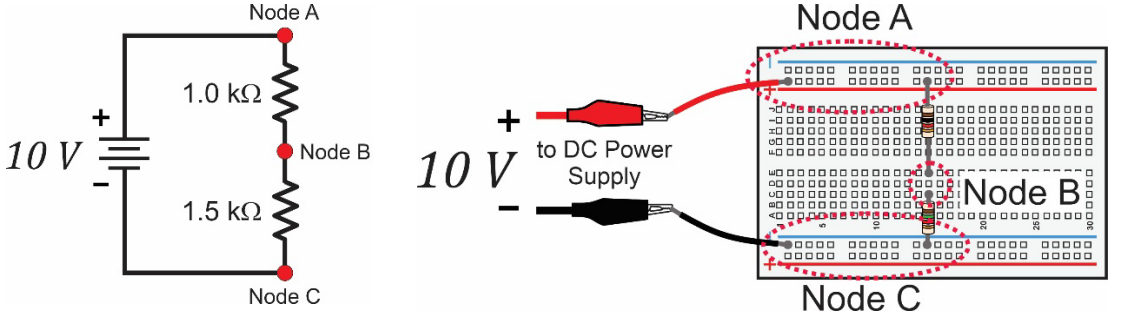


(a) (b)

Figure 3.4 Voltage-divider circuit: (a) schematic and (b) breadboard implementation

3.5 Identify the Circuit Nodes

ACTION: Examine Figure 3.5 and identify **Node A**, **Node B**, and **Node C** in your circuit. These nodes will be used as reference points for the voltage measurements.



(a) (b)

Figure 3.5 Voltage-divider circuit with Nodes A, B, and C identified: (a) schematic and (b) breadboard implementation.

4.0 Measuring Voltage with the DMM

In this section, the DMM will be configured as a DC voltmeter to measure voltage between nodes in the voltage-divider circuit. Because voltage is measured between two points, the DMM is connected in parallel with the circuit element or node pair being measured.

4.1 Configure the DMM to Measure DC Voltage

ACTION: Press the **DCV** button, as shown in Figure 4.1. Verify that **DCV** appears on the display.

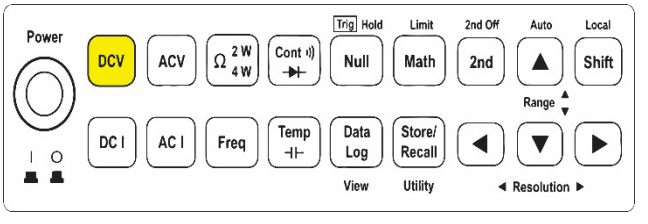


Figure 4.1 DMM front-panel controls with the DCV button highlighted.

4.2 Connect the DMM Test Leads

ACTION: Verify that the red test lead is connected to the **V/Ω HI** input and the black test lead is connected to **LO**, as shown in Figure 4.2. Use short jumper wires with the alligator clips when connecting to breadboard nodes.

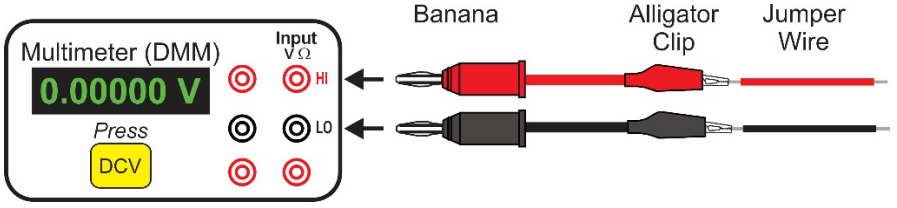


Figure 4.2 DMM test-lead and jumper-wire connections for measuring DC voltage

4.3 Measure the DC Voltage Across Nodes A and C

ACTION: Connect the red DMM lead to Node A and the black DMM lead to Node C, as shown in Figure 4.3. Turn on the selected power-supply channel, then measure and record V_{AC}.

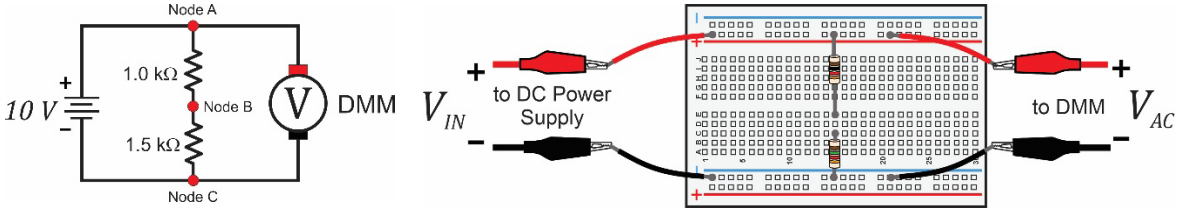


Figure 4.3 DMM connected across Nodes A and C: (a) schematic and (b) breadboard implementation.

QUESTIONS: On the handout, write your answers to the questions listed under Section 4.3

4.4 Observe the Effect of Reversing the DMM Leads

ACTION: Reverse the DMM leads so that the red lead is connected to Node C and the black lead is connected to Node A, as shown in Figure 4.4. Record V_{CA}.

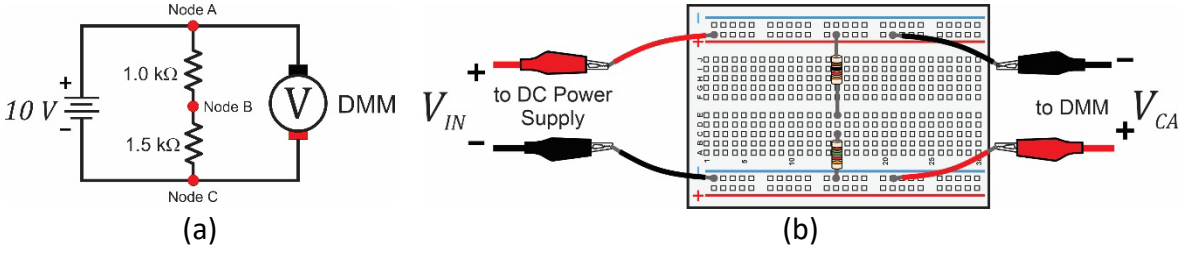
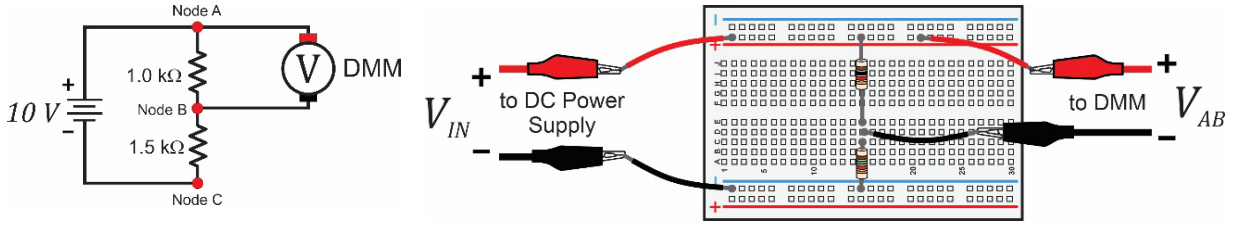


Figure 4.4 DMM connected across Nodes C and A with reversed polarity: (a) schematic and (b) breadboard implementation.

QUESTIONS: On the handout, write your answers to the questions listed under Section 4.4

4.5 Measure the Voltage Across R_1

ACTION: Connect the red DMM lead to Node A and the black DMM lead to Node B, as shown in Figure 4.5. Measure and record V_{AB}, the voltage across R_1.



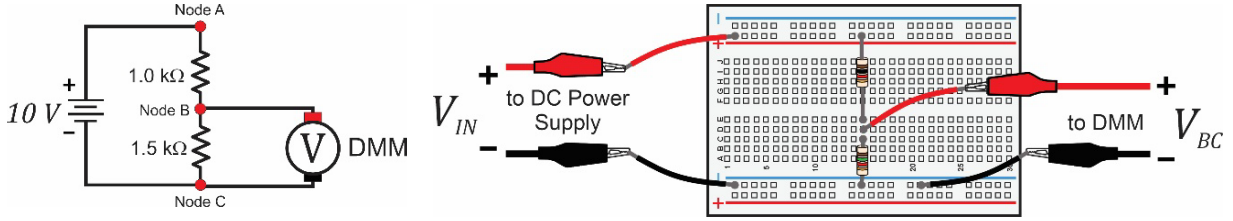
(a) (b)

Figure 4.5 Measurement of V_{AB} across R_1: (a) schematic and (b) breadboard implementation.

QUESTIONS: On the handout, write your answers to the questions listed under Section 4.5

4.6 Measure the Voltage Across R_2

ACTION: Connect the red DMM lead to Node B and the black DMM lead to Node C, as shown in Figure 4.6. Measure and record V_{BC}, the voltage across R_2



(a) (b)

Figure 4.6 Measurement of V_{BC}across R_2: (a) schematic and (b) breadboard implementation.

QUESTIONS: On the handout, write your answers to the questions listed under Section 4.6

4.7 Compare Measured Voltages to Voltage-Divider Calculations

ACTION: Using the measured values of R_1, R_2, and $V_{AC} = V_{IN}$, calculate the expected voltage across each resistor:

$$V_1 = V_{IN}\frac{R_1}{(R_1 + R_2)}$$

$$V_2 = V_{IN}\frac{R_2}{(R_1 + R_2)}$$

NOTE: Using the measured values of R_1, R_2, and $V_{AC} = V_{IN}$, calculate the expected voltage across each resistor:

QUESTIONS: On the handout, write your answers to the questions listed under Section 4.7

5.0 Measuring Current with the DMM

In this section, the DMM will be configured as an ammeter and inserted in series with the voltage-divider circuit. Because the circuit is a single series path, the same current flows through R_1, R_2, and the power supply.

SAFETY: Never connect the DMM across the power supply or across a resistor while it is configured as an ammeter. This creates a low-resistance path that may blow the DMM fuse, damage the meter, or short-circuit a component.

5.1 Configure the DMM to Measure Current

ACTION: Press the **DCI** button, as shown in Figure 5.1. Verify that **DCI** appears on the display.

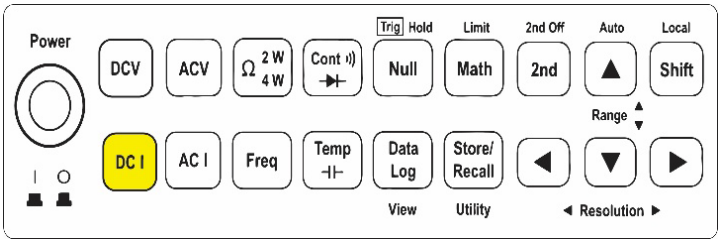


Figure 5.1 DMM front-panel controls with the DCI button highlighted.

5.2 Test Lead Connections to the DMM

ACTION: Move the red test lead from the **V/Ω HI** input to the **100 mA** input. Leave the black test lead connected to **LO**, as shown in Figure 5.2. Continue using short jumper wires to connect the test leads to the breadboard.

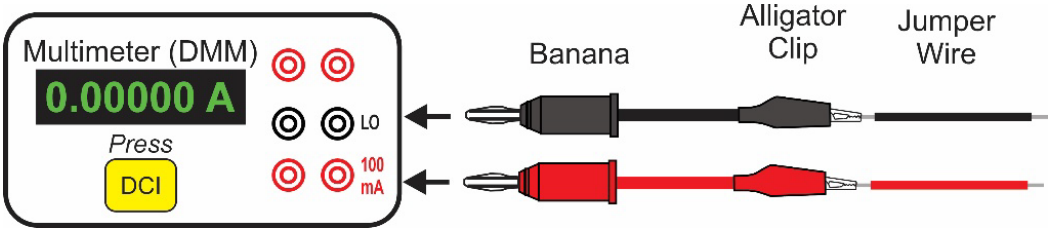


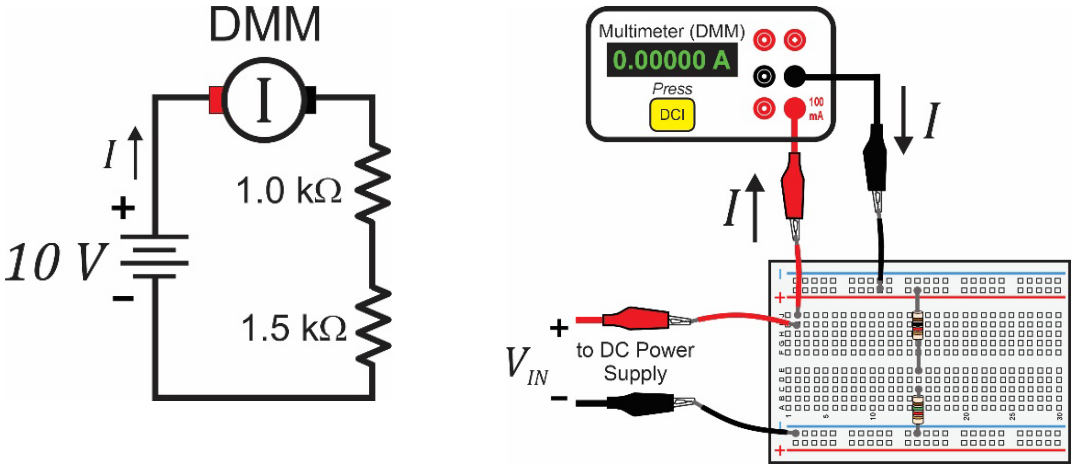
Figure 5.2 DMM test-lead and jumper-wire connections for measuring DC current

5.3 Insert the DMM Ammeter into the Voltage-Divider Circuit

ACTION: Turn off the power-supply channel. Insert the DMM in series between the power supply and R_1, as shown in Figure 5.3.

SAFETY: Before turning the channel back on, verify that the DMM is connected in series. Never connect the DMM across the power supply or either resistor while it is configured as an ammeter.

NOTE: Current entering the red **100 mA** terminal produces a positive reading. A negative reading indicates that the current is entering the black **LO** termina.

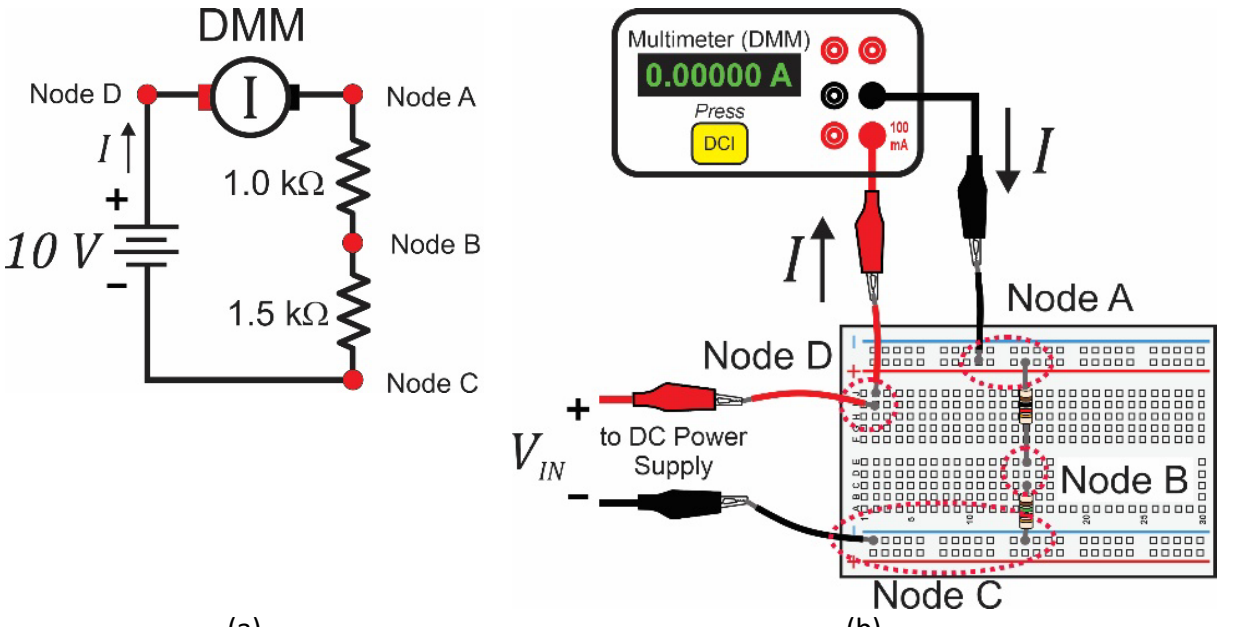


(a) (b)

Figure 5.3 DMM inserted in series with the voltage-divider circuit: (a) schematic and (b) breadboard implementation.

5.4 Identify the Circuit Nodes

ACTION: Identify Nodes A, B, C, and D using Figure 5.4. Inserting the ammeter opens the connection between the power supply and R_1, creating Node D on the supply side and Node A on the resistor side.



(a) (b)

Figure 5.4 Voltage-divider circuit with the DMM inserted as an ammeter and Nodes A, B, C, and D identified: (a) schematic and (b) breadboard implementation.

5.5 Measure the Circuit Current

ACTION: Turn on the power-supply channel. With the DMM Ammeter connected in series, measure and record the circuit current I.

QUESTIONS: On the handout, write your answers to the questions listed under Section 5.5

6.0 Verify That Current Is the Same Throughout a Series Circuit

ACTION: Turn off the power-supply channel. Move the DMM to another location in the series path, such as between R_1 and R_2, as shown in Figure 6.0. Turn the channel back on and record the current.

SAFETY: Before turning on the channel, verify that the DMM is configured as an ammeter and connected in series.

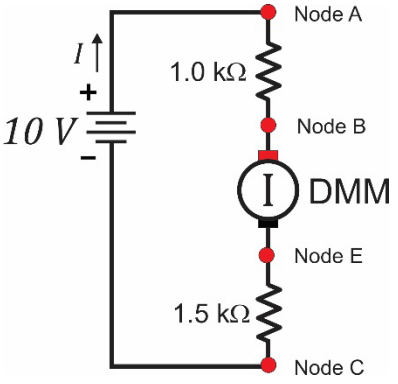


Figure 6.0 DMM inserted between R_1 and R_2 to measure current at a second location.

QUESTIONS: On the handout, write your answers to the questions listed under Section 6.0

7.0 Lab Complete

LAB COMPLETE: Turn off the power-supply channel output and disconnect the DMM and power-supply leads. Return the test cables to the cart, and leave the breadboard at your workstation. Leave the equipment turned on.